2.4 Sum and Difference Formulas

Name _____ Period ____ Date _____

Pre-Calculus 2.4 · Student sheet

I can…

- use sum and difference formulas to find exact values.
- use sum and difference formulas to rewrite and evaluate expressions.
- use sum and difference formulas to solve equations.

Vocabulary · fill in the missing word

Sum and difference formulas — $\sin(u \pm v) =$ _____; $\cos(u \pm v) = \cos u \cos v \mp \sin u \sin v$; $\tan(u \pm v) = (\tan u \pm \tan v)/(1 \mp \tan u \tan v)$

Part 1 · Does it distribute?

Use a calculator in degree mode. Round to three decimal places.

Expression	Value
$\cos(30° + 60°)$	
$\cos 30° + \cos 60°$	
$\cos 30° \cos 60° - \sin 30° \sin 60°$	
$\sin(45° + 45°)$	
$\sin 45° + \sin 45°$	
$\sin 45° \cos 45° + \cos 45° \sin 45°$	

Does $\cos(u + v) = \cos u + \cos v$? _____ Which expression matched $\cos(30° + 60°)$? _____

Part 2 · Notes: fill in the blanks

$\sin(u + v) =$ _____ $\sin(u - v) =$ _____

$\cos(u + v) =$ _____ $\cos(u - v) =$ _____

$\tan(u + v) =$ _____ $\tan(u - v) =$ _____

In the sine formulas the sign _____; in the cosine formulas the sign _____.

Part 3 · Examples (work them with the class)

Example 1

Find the exact value of $\sin 75°$.

Example 2

Find the exact value of $\cos(\frac{\pi}{12})$.

Example 3

Find the exact value of $\tan 15°$.

Example 4

Write $\sin 42° \cos 12° - \cos 42° \sin 12°$ as a trigonometric function of one angle. Then find its exact value.

Example 5

$\sin u = \frac{5}{13}$ with $0 < u < \frac{\pi}{2}$, and $\cos v = -\frac{3}{5}$ with $\frac{\pi}{2} < v < \pi$. Find the exact value of $\sin(u + v)$.

Example 6

Find all solutions in $[0, 2\pi)$: $\sin(x + \frac{\pi}{4}) + \sin(x - \frac{\pi}{4}) = 1$

Part 4 · Partner practice

Partner 1

Find the exact value of $\cos 105°$.

Partner 2

If $\tan u = \frac{1}{2}$ and $\tan v = \frac{1}{3}$, find $\tan(u + v)$.